

# Foundations of Large Language Models: What They Are and How They Behave

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# Lecture 1

# A Familiar Shift in a New Domain

We are experiencing a familiar technological shift in a new cognitive domain.

- ▶ Calculators reduced the need for manual arithmetic but not mathematical understanding.
- ▶ Students now ask the same question in a new form: why learn this if AI can do it.
- ▶ The domain of external assistance has shifted from computation to language and reasoning.
- ▶ External tools can perform tasks without replacing the need for internal competence.
- ▶ Real-world situations still require unassisted thinking, speaking, and judgment.
- ▶ The analogy is useful only if we understand what LLMs actually are and are not.

# What LLMs Fundamentally Are

Large language models are best understood as systems for sequence transformation.

- ▶ At a fundamental level, LLMs map sequences of tokens to new sequences.
- ▶ They are trained on large corpora such as Common Crawl and other aggregated text sources.
- ▶ Their objective is next-token prediction rather than truth or correctness.
- ▶ Outputs are conditioned entirely on the provided context and prior generated tokens.
- ▶ They do not contain structured knowledge in the manner of databases or symbolic systems.
- ▶ Their behavior is better described as transformation than as understanding or reasoning.

# How Context Shapes Output

The structure and content of the input largely determine the direction of the output.

- ▶ The model responds to patterns in the prompt rather than to an internal world model.
- ▶ Small changes in phrasing or framing can substantially alter the generated response.
- ▶ Structured input constrains the model toward more stable and predictable outputs.
- ▶ Ungrounded prompts invite unconstrained sampling from the training distribution.
- ▶ Context functions simultaneously as instruction and as constraint on generation.
- ▶ Effective use requires deliberate construction of input rather than passive querying.

# Why LLMs Perform Well

LLMs are effective when tasks can be framed as transformations between representations.

- ▶ Strong performance appears when translating between formats such as text, schema, and code.
- ▶ The models excel at recognizing and reproducing common structural patterns.
- ▶ Tasks with clear mappings between representations are especially reliable.
- ▶ Performance depends on statistical regularities learned from large training corpora.
- ▶ Structured prompts reduce ambiguity and improve consistency of the output.
- ▶ This explains why some tasks appear intelligent while others degrade very quickly.

# Evidence from Structured Tasks

Empirical evidence shows that reliability increases when work is staged through explicit structure.

- ▶ In structured cases such as status effects, schema to JSON to code transitions were accurate.
- ▶ Consistent field definitions enabled alignment across multiple representations.
- ▶ Unit tests generated from structured inputs were valid and executable.
- ▶ Minimal correction was needed when scaffolding and constraints were present.
- ▶ The model maintained coherence across several transformation steps.
- ▶ Reliability therefore appears to be conditional on structure rather than inherent capability.

# Failure Mode 1: Hallucination and Fabrication

When grounding is weak, LLMs often produce plausible statements that are false.

- ▶ Models may invent facts, citations, or numerical values that appear credible.
- ▶ In the EDA experiment, trends were inferred from minimal data and presented as factual.
- ▶ Fabricated regression test outputs were produced without any actual computation.
- ▶ Hallucinations are patterned and plausible rather than purely random.
- ▶ Users often accept such outputs because fluency is mistaken for reliability.
- ▶ This creates systematic risk in research, analysis, and decision-making work.

# Failure Mode 2: Structural vs Semantic Errors

LLMs often preserve formal structure while losing semantic correctness.

- ▶ Code may compile or appear correct while implementing the wrong logic.
- ▶ In the dice example, probabilistic reasoning was replaced by a uniform distribution.
- ▶ Structural patterns are easier for the model to reproduce than domain-specific meaning.
- ▶ The model can preserve form without validating function.
- ▶ Such errors often remain hidden unless explicit tests are applied.
- ▶ Correct structure therefore does not imply correct semantics.

# Failure Mode 3: Context Drift

Long interactions often degrade because context accumulates noise and irrelevant patterning.

- ▶ Extended conversations accumulate unintended context that affects later outputs.
- ▶ Models may introduce irrelevant artifacts such as code blocks in unrelated tasks.
- ▶ Output style and format may drift unless actively corrected.
- ▶ Earlier errors can propagate and distort future responses.
- ▶ The model does not maintain clean topic separation unless the user enforces it.
- ▶ Managing context is therefore an active responsibility rather than a passive assumption.

# The Core Limitation

The central limitation of LLMs is that they generate by constrained pattern completion rather than truth-tracking.

- ▶ There is no internal mechanism that independently verifies correctness.
- ▶ The model cannot on its own validate facts, proofs, or computations.
- ▶ It lacks persistent state and stable internal representations across interactions.
- ▶ It does not reliably track contradictions unless prompted and constrained to do so.
- ▶ Reliability must therefore be imposed externally through structure and validation.
- ▶ This limitation is fundamental to the system rather than a temporary defect.

# Implications for Users

Effective use of LLMs requires disciplined control over inputs, outputs, and validation.

- ▶ Begin with grounded inputs such as documents, data, or explicit references.
- ▶ Treat outputs as drafts or hypotheses rather than as final answers.
- ▶ Correct the model explicitly when it deviates from the task.
- ▶ Avoid copying generated text without careful inspection and revision.
- ▶ Use iteration, comparison, and cross-checking to improve reliability.
- ▶ The model interprets silence as acceptance and reinforces that interaction pattern.

# Discussion

Understanding LLM behavior leads directly to the problem of disciplined use.

- ▶ LLMs are powerful systems but they are inherently unreliable when unconstrained.
- ▶ Their strongest performance depends on structure, staging, and explicit constraints.
- ▶ Unstructured use leads predictably to degraded outputs and poor reasoning outcomes.
- ▶ Effective use is therefore a matter of interaction design rather than mere access.
- ▶ The real challenge is not possession of the tool but disciplined integration of it.
- ▶ This leads to the next question: how can we use these systems without losing rigor.

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